NASA SPACE APPS CHALLENGE // MISSION BRIEFING

Planetary Analogue Finder

Satellite-driven geospatial intelligence platform identifying terrestrial testbeds for Lunar and Martian surface operations.

01 // MISSION CHALLENGE

High-Stakes Extraterrestrial Hardware Validation

Zero Margin for Field Failure

Deploying rovers, drills, and EVA suits to Lunar craters or Martian plains requires prior terrestrial validation under matching slope, wind, and regolith constraints.

- **Terrain Steepness:** Wheel traction limits on crater walls.
- **Regolith Composition:** Basaltic ash versus iron-oxide dust.
- **Solar Irradiance:** Power generation at high latitudes.
- **Wind Loadings:** Aeolian erosion and dust storm simulation.

02 // SYSTEM ENGINE

Multi-Factor Geospatial Telemetry Matching

5 VECTORS

Real-time weighted scoring across elevation, terrain slope, wind velocity, solar radiation, and mineral regolith classification to pinpoint optimal Earth analogues.

03 // OPERATIONAL PIPELINE

End-to-End Target Site Selection

STEP 01

Target Body

Select Moon, Mars, or dual-mission profiles.

STEP 02

Query & Sliders

Input natural-language specs and telemetry limits.

STEP 03

Vector Ranking

Evaluate percentage match per environmental factor.

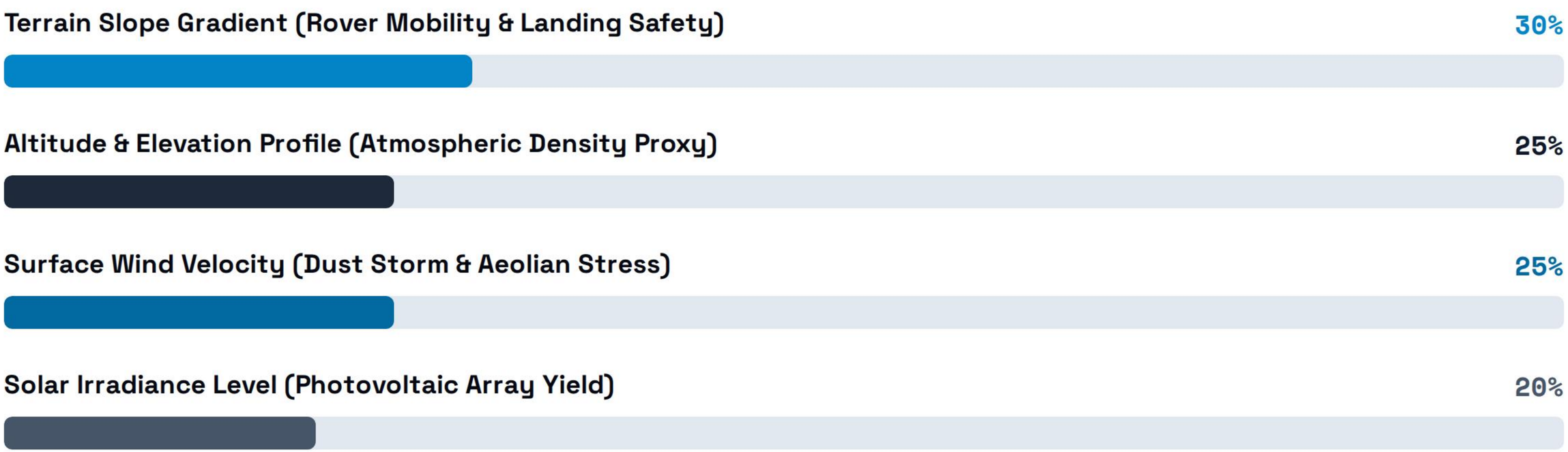
STEP 04

Orbital Export

Inspect satellite imagery and export GPS CSV.

04 // SCORING ALGORITHM

Environmental Similarity Weight Distribution



05 // VERIFIED FIELD SITES

Terrestrial Testbeds & Extraterrestrial Counterparts

Terrestrial Analogue	Planetary Counterpart	Primary Validation Role
Atacama Desert, Chile	**Mars: Jezero Crater**	Hyper-arid perchlorate soil & biosignature sensors
Devon Island, Canada	**Moon: Shackleton Crater**	Polar impact crater EVA suit & logistics drills
Mauna Kea, Hawaii	**Moon: Mare Tranquillitatis**	High-slope volcanic basaltic regolith traction
McMurdo Dry Valleys	**Mars: Utopia Planitia**	Sub-zero permafrost autonomous rover navigation

06 // SYSTEM STACK

Cloud Deployment & Orbital Visualization

Python & Streamlit

Instant cloud execution via Streamlit Community Cloud.

Esri Satellite Layers

High-resolution orbital imagery toggle on Folium maps.

Telemetry CSV Export

Downloadable GPS coordinates and similarity scores.

READY FOR SURFACE DEPLOYMENT

Advancing Humanity’s Next Giant Leap

Live interactive prototype deployed on Streamlit Community Cloud for NASA Space Apps Challenge evaluation.